

Mathematics Analysis and Approaches HL Exploration

Aim: Exploring the strengths and limitations of Separating Variables, Laplace transform and Variational Calculus methods to solve a partial differential equation that models the phenomenon of conduction in a one dimensional metal rod.

1.0 Introduction

Heat transfer is a fundamental process that governs a wide range of physical and engineering phenomena right from the cooling of electronic devices to the thermal regulation of buildings and even biological heat distribution in living organisms. The heat equation is a second-order partial differential equation and is one of the most essential models in mathematical physics, describing how temperature evolves over time in a given medium due to thermal conduction. This equation is derived from Fourier's law of heat conduction and the principle of energy conservation, making it a crucial tool for understanding temperature variations in solids, fluids, and even astrophysical bodies.

In this exploration I aim to analyze the one dimensional heat equation in the context of a thermally conductive rod with a distributed internal heat source. Such a scenario is highly relevant in practical applications, such as heat dissipation in electrical circuits, the thermal behavior of metal bars in manufacturing and the diffusion of heat in planetary crusts. The presence of an internal heat source that I chose to specifically to increase the complexity of this problem makes analytical solutions challenging thus necessitating the use of various mathematical techniques to obtain accurate temperature distributions.

By applying multiple solution approaches including separation of variables Laplace transforms and the variational method I aim to explore the strengths and limitations of different mathematical frameworks for solving a given problem. I aim to uncover the advantages and disadvantages of each method where in brief the separation of variables decomposes the solution into steady-state and transient components, Laplace transforms convert the problem into an algebraic equation for easier manipulation, and variational techniques optimize the solution by depicting the impact of small variations build the final solution. By comparing these approaches, I aim to gain deeper insights into mathematical problem solving in physics and how different methods were innovated in order to adapt to complex boundary conditions and forcing terms that mimic real world phenomena.

2.0 Defining Variables for Different Methods

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} + S_0 \sin\left(\frac{\pi x}{L}\right)$$

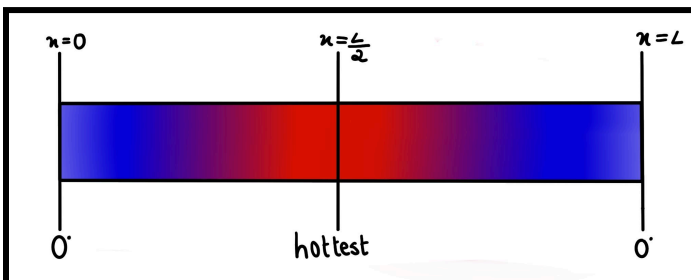
General	
t	Time variable that our heat function is dependent on where $t \geq 0$
x	The distance variable that is defined as the distance of a point on the rod from the left extreme end ranging from $0 \leq x \leq L$ where L is the length of the rod
$u(x, t)$	Function that depicts the distribution of heat and in turn the temperature gradient in a one dimensional conductive rod. The temperature variation function is dependent on

	both time and distance and describes this distribution for a moment in time across the length of the rod.
α	Conductivity constant of the rod that depends on the material of the rod.
$S_0 \sin(\frac{\pi x}{L})$	Sinusoidal heat source mathematically known as the forcing term that radiates heat from the centre towards both ends of the rod while it's intensity varies with x .
$f(x)$	Initial distribution of heat and temperature gradient. That is when $u(x, 0)$
Separating the Variables	
$u_s(x)$	Steady State solution that determines what happens to the function as $t \rightarrow \infty$
$w(x, t)$	Transient solution that remains after the steady state solution reaches ∞
C	All capital C's are integration constants
$X(x)$	A function only dependent on x
$T(t)$	A function only dependent on t
λ	Eigenvalue that we consider to split the time and distance variables in order to decompose the partial differential equation into two ordinary differential equations.
k	Is the particular solution constant
Laplace Transform Method	
s and p	Laplace transform Variable Constant
$L\{\}$	Represents laplace transform
w	Is the function that represents the homogeneous solution when the forcing term is considered to be 0. This is the same as the transient function in Separating the Variables method
W	Is the transformed transient function after applying laplace transform to both sides of the function. Note that w

	transforms to W .
$u_s(x)$	Is the steady state solution in this method as well
$U_s(p)$	Is the transformed steady state solution similar to W . Upon applying laplace transform $u_s(x)$ transforms to $U_s(p)$.
Variation Method	
$v(x)$	This is the test function where the integral of the production of the test function and $u(x, t)$ satisfies the test function.
$\phi_n(x)$	Laplacian Eigenfunction that satisfies $u(x, t)$
$c_n(t)$	Time dependent coefficients that satisfy $u(x, t)$
$-\lambda_n$	Is the eigenvalue that satisfies the eigenfunction chosen
IF	Integrating factor whose formula is given by $IF = e^{\int p(x)dx}$ where $p(x)$ is found in the differential equation in the form $\frac{dy}{dx} + p(x)y = q(x)$

3.0 Methods to solve

3.1 Defining boundary and initial conditions



3.1.1 Boundary conditions are as follows:

$$u(0, t) = u(L, t) = 0$$

These conditions are known as the Dirichlet conditions which implies, in the context of the chosen equation, that both ends of the conducting rod are assumed to be kept at a constant temperature and this temperature is 0. This is done by keeping

the rod in contact with a cool reservoir that absorbs all heat. Mathematically I chose these boundary conditions as they heavily simplify the calculation of integration constants that are essentially only translations to the obtained function that can be varied later. This will allow us to obtain a general equation that can model more complex scenarios.

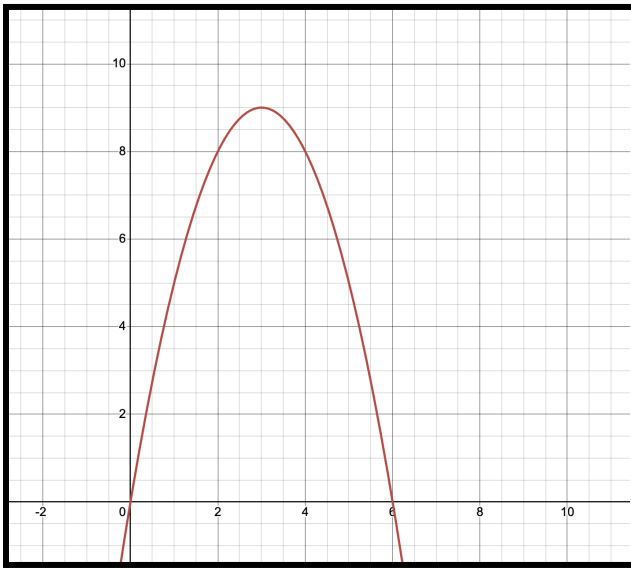
3.1.2 Initial conditions are as follows:

$$u(x, 0) = f(x) = x(L - x)$$

This distribution can be explained using Figure 1.0 and Graph 1.0 that shows that initially heat is distributed in the form of a quadratic function and the vertex defines maximum heat at the centre of the rod.

Figure 1.0

This heat decreases exponentially from both sides of the centre of the rod reaching 0 at the extreme ends. The colour gradient in Figure 1.0 depicts this exponential distribution where red means hottest and blue means coolest.



Graph 1.0 depicts the initial heat distribution in the rod as a quadratic equation. It's roots imply that the temperature at the ends of the rod is 0 and its vertex implies maximum temperature at $\frac{L}{2}$

3.2 Separating the variables analytical method:

The equation we are considering has two derivative elements, one with respect to time and the other with respect to distance along the rod from origin. This implies that our function u is a partial

differential equation that is given to be dependent on derivatives of different variables.

Moreover the equation we are considering for this exploration poses to be a slightly more complex problem due to the presence of the internal heating sinusoidal component that can be seen as $S_0 \sin(\frac{\pi x}{L})$ in the equation. Since this component of the problem is not dependent on the derivative of time or displacement on the rod the equation is a non-homogeneous partial differential equation.

Hence we need to start off by splitting the problem into two parts:

- 1) Find a steady state solution, given by $u_s(x)$, that satisfies the non-homogeneous equation when $\frac{\partial u}{\partial t} = 0$. This solution will give us the time independent function that determines how heat is distributed when at equilibrium.
- 2) Then we solve the homogeneous equation for the temporal part of the differential equation, given by $w(x, t)$, which describes how the system evolves from the initial condition to the steady state condition.

Therefore the solution will decompose and be given by:

$$u(x, t) = u_s(x) + w(x, t)$$

3.2.1 Finding the Steady State solution to determine $u_s(x)$:

At steady state we assume no time dependence, therefore $\frac{\partial u}{\partial t} = 0$ thereby simplifying the equation to:

$$\alpha \frac{\partial^2 u_s}{\partial x^2} + S_0 \sin\left(\frac{\pi x}{L}\right) = 0$$

Dividing by α gives:

$$\frac{\partial^2 u_s}{\partial x^2} = -\frac{S_0}{\alpha} \sin\left(\frac{\pi x}{L}\right)$$

Since we only have one variable in the differential equation we can simply integrate twice to reduce the $\frac{\partial^2 u_s}{\partial x^2}$ term to u_s

$$\int \int \frac{\partial^2 u_s}{\partial x^2} = \int \left[-\frac{S_0}{\alpha} \sin\left(\frac{\pi x}{L}\right) \right] dx$$

$$\int \frac{\partial u_s}{\partial x} = \int \frac{S_0}{\alpha} \frac{L}{\pi} \cos\left(\frac{\pi x}{L}\right) + C_1 dx$$

$$u_s = \frac{S_0}{\alpha} \left(\frac{L}{\pi}\right)^2 \sin\left(\frac{\pi x}{L}\right) + C_1 x + C_2$$

Now we can apply the previously defined boundary conditions to find C_1 and C_2 by substituting in the function u_s :

We know:

$$u_s(0) = 0 \text{ and } u_s(L) = 0$$

Therefore at $x = 0$

$$0 = \frac{S_0}{\alpha} \left(\frac{L}{\pi}\right)^2 \sin(0) + C_1 0 + C_2$$

$$C_2 = 0$$

At $x = L$

$$0 = \frac{S_0}{\alpha} \left(\frac{L}{\pi}\right)^2 \sin(\pi) + C_1 L$$

Since:

$$\sin(\pi) = 0$$

Then:

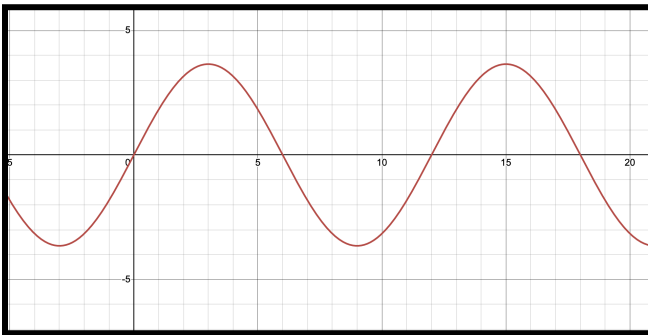
$$C_1 L = 0$$

Hence:

$$C_1 = 0$$

Therefore the steady state solution is:

$$u_s = \frac{S_0}{\alpha} \left(\frac{L}{\pi}\right)^2 \sin\left(\frac{\pi x}{L}\right)$$



This solution implies that the long term temperature distribution when all transient effects have disappeared takes the form of a sinusoidal wave and surprisingly restores the initial distribution but with a higher peak at $\frac{L}{2}$ due to the heat emitted by the source. This equations gives an important relationship between the peak temperature and S_0 .

Graph 2.0 depicts the solution function

3.2.2 Finding the time dependent solution $w(x, t)$:

Now we must find the solution that depends on time while understanding how and to what extent time can affect heat distribution in the conductive rod which is the crux of this exploration. We can do this by simplifying the differential equation and assuming that the internal heat source is equal to zero and solving the homogeneous equation we get to find $w(x, t)$:

$$\frac{\partial w}{\partial t} = \alpha \frac{\partial^2 w}{\partial x^2}$$

We now must find a way to convert the partial differential equation into an ordinary differential equation. If we assume the solution to be made up for two functions where x and t are independent of each other we can split the PDE into two ODE. Hence we will assume that:

$$w(x, t) = X(x)T(t)$$

Note: $X(x)$ is referred to as X and $T(t)$ is referred to as T further. The purpose behind using $X(x)$ and $T(t)$ is to make it explicit that both the functions depend only on one variable.

Substituting this into the PDE

$$X \frac{dT}{dt} = \alpha \frac{d^2 X}{dx^2} T$$

Dividing by XT

$$\frac{1}{T} \cdot \frac{dT}{dt} = \alpha \frac{1}{X} \cdot \frac{d^2 X}{dx^2}$$

The purpose of the above steps can be made explicit by understanding the above equation. By splitting the function $w(x, t)$ into independent functions of x and t that we are calling $X(x)$ and $T(t)$, we have in the above equation, the LHS which is independent of x and only dependent on time and the RHS which is only dependent on x and independent of time. Therefore since x and t are independent variables they must be equal to the same constant that we will consider to be a non zero value — λ . This is known as an eigenvalue.

The reason behind choosing a negative eigenvalue will be made explicit in Section 3.2.4

$$\frac{1}{\alpha T} \cdot \frac{dT}{dt} = \frac{1}{X} \cdot \frac{d^2 X}{dx^2} = -\lambda$$

By doing this we get two ordinary differential equations:

1) Spatial equation

$$\frac{1}{X} \cdot \frac{d^2 X}{dx^2} = -\lambda$$

2) Temporal equation

$$\frac{1}{\alpha T} \cdot \frac{dT}{dt} = -\lambda$$

3.2.3 Solving for the Spatial Function $X(x)$:

Now that we have obtained the above ordinary differential equations we can solve them individually. Since the differential equation was split using an eigenvalue we will have to use method of hit and trial solution to find the value of λ in terms of defined variables.

Therefore using the method of trial solution

$$\frac{d^2 X}{dx^2} + \lambda X = 0$$

Let's assume that a particular solution in the form

$$X(x) = ae^{bkx+c}$$

Justification: We have assumed a and b as 1 and c to be zero to find the fundamental equation. Furthermore the boundary conditions and assumptions made so far will not be sufficient to determine these constants. Therefore our particular solution will be:

$$X(x) = e^{kx}$$

Substitute X_c in homogeneous differential equation

$$k^2 e^{kx} + \lambda e^{kx} = 0$$

Since $e^{kx} \neq 0$

We get

$$k^2 = -\lambda$$

$$k = \pm i\sqrt{\lambda}$$

For these complex roots the solution X_c can be written in trigonometric form using Euler's formula where:

$$e^{i\sqrt{\lambda}x} = \cos(\sqrt{\lambda}x) + i\sin(\sqrt{\lambda}x)$$

as:

$$X(x) = A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}x)$$

Solving to find A and B by applying boundary conditions:

$$1) \quad X(0) = 0$$

$$0 = A\cos(0) + B\sin(0)$$

$$A = 0$$

$$2) \quad X(L) = 0$$

$$0 = B\sin(L\sqrt{\lambda})$$

For a non-trivial solution where $B \neq 0$ we need

$$\sin(L\sqrt{\lambda}) = 0$$

to satisfy the equation.

This is only possible when

$$L\sqrt{\lambda} = n\pi \quad \text{where } n = 1, 2, 3, \dots$$

So,

$$\lambda_n = \frac{n^2 \pi^2}{L^2}$$

Therefore the general solution where $A = 0$ can be found by substitution $\lambda_n X_c$ to give $X(x)$:

$$X_n(x) = B_n \sin\left(\frac{n\pi x}{L}\right)$$

3.2.4 solving for the Temporal Function $T(t)$:

$$\frac{dT}{dt} = -\alpha \lambda T$$

Note that $T(t)$ is dependent on x

Substituting λ_n gives

$$\frac{dT_n}{dt} = -\alpha \frac{n^2 \pi^2}{L^2} T_n$$

Using the method of trial solution

Let's assume that a particular solution in the form

$$T = e^{kt}$$

Substitute T in homogeneous differential equation

$$ke^{kt} = -\alpha \frac{n^2 \pi^2}{L^2} e^{kt}$$

Dividing by e^{kt} since $e^{kt} \neq 0$

$$k = -\alpha \frac{n^2 \pi^2}{L^2}$$

Therefore considering the above solution for k we get $T_n(t)$ as

$$T_n(t) = e^{-\alpha \frac{n^2 \pi^2}{L^2} t}$$

This is where we can understand the importance of taking the eigenvalue negative. If we had written the separation

constant as λ instead of $-\lambda$ then our solution would be $T_n(t) = e^{\alpha \frac{n^2 \pi^2}{L^2} t}$ implying that the solution exponentially grows with time which is physically incorrect in the chosen heat distribution problem thus violating the second law of thermodynamics. Heat must dissipate over time and not spontaneously increase since no external energy is added. The solution we found however, considering the eigenvalue to be negative shows the expected decrease in temperature over time due to dissipation of heat.

3.2.5 Compiling complete solution for $w(x, t)$:

Since the solution exists for all values of n , the general solution will be the sum of all possible solutions due to the principle of superposition. Since the equation found is linear then the sum of all individual solutions is a valid solution. We will consider the fundamental solution to be the mode i.e. when $n=1$. This gives us the specific spatial pattern when $n=1$ where

- The function $\sin(\frac{n\pi x}{L})$ represents a standing wave in space, where n determines the number of oscillations along the length of the system.
- The time-dependent factor $e^{-\alpha \frac{n^2 \pi^2}{L^2} t}$ dictates how fast each mode decays over time due to heat diffusion. Higher modes ($n=2,3$) decay faster than lower modes, leaving only the fundamental mode ($n=1$) as $t \rightarrow \infty$.

Since each mode satisfies the heat equation independently, we can construct the total solution as the superposition or sum of all possible modes. Initially, many modes contribute to the solution, but over time, higher-order modes fade away, leaving only the dominant, slowest-decaying mode, which determines the steady-state behavior.

Therefore:

$$w(x, t) = \sum_{n=1}^{\infty} B_n e^{-\alpha \frac{n^2 \pi^2}{L^2} t} \sin(\frac{n\pi x}{L})$$

3.2.6 Determining B_n :

We can determine B_n through the use of the initial condition where

$$u(x, 0) = f(x)$$

Therefore substituting $t = 0$

$$w(x, 0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

Therefore considering the initial conditions equation:

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

Multiple both sides by $\sin\left(\frac{m\pi x}{L}\right)$ to use orthogonality to eliminate other terms (Explained below)

$$f(x) \cdot \sin\left(\frac{m\pi x}{L}\right) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right)$$

Integrate both sides from 0 to L

$$\int_0^L f(x) \cdot \sin\left(\frac{m\pi x}{L}\right) = \sum_{n=1}^{\infty} \int_0^L B_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right)$$

Orthogonality is a mathematical property for perpendicular functions. In the context of this equation, orthogonality will allow us to isolate each coefficient B_n in the Fourier series expansion of the initial conditions $f(x)$. By multiplying both sides of the equation by $\sin\left(\frac{m\pi x}{L}\right)$ and integrating over the domain, we take advantage of the fact that sine functions of different modes are orthogonal meaning their integral over the interval is zero unless they are the same mode. This ensures that when we integrate, all terms in the infinite sum vanish except for the one where $m = n$, allowing us to explicitly solve for B_n in terms of n .

- If $m \neq n$ the solution to the integral $\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right)$ is 0
- But if $m = n$ then the solution to the integral is a non-zero number.

Therefore we will assume that $m = n$ and solve the integral on the LHS

$$\begin{aligned} & \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) \\ &= \frac{1}{2} \int_0^L 1 - \cos 2\left(\frac{n\pi x}{L}\right) \\ &= \frac{1}{2} L - \int_0^L \cos 2\left(\frac{n\pi x}{L}\right) \\ &= \frac{1}{2} L - \left[\frac{L}{2n\pi} \sin 2\left(\frac{n\pi x}{L}\right) \right]_0^L \\ &= \frac{L}{2} \end{aligned}$$

Now substitute this value giving

$$\begin{aligned} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) &= B_n \frac{L}{2} \\ B_n &= \frac{2}{L} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) \end{aligned}$$

Substitute $f(x)$:

$$B_n = \frac{2}{L} \int_0^L x(L - x) \cdot \sin\left(\frac{n\pi x}{L}\right)$$

Splitting into two integrals by expanding $x(L - x)$

$$B_n = \frac{2}{L} \int_0^L xL \cdot \sin\left(\frac{n\pi x}{L}\right) - \frac{2}{L} \int_0^L x^2 \cdot \sin\left(\frac{n\pi x}{L}\right)$$

Simplifying by removing constants out of the integral

$$B_n = 2 \int_0^L x \cdot \sin\left(\frac{n\pi x}{L}\right) - \frac{2}{L} \int_0^L x^2 \cdot \sin\left(\frac{n\pi x}{L}\right)$$

Solve both integrals using integration by parts beginning with $2 \int_0^L x \cdot \sin\left(\frac{n\pi x}{L}\right)$

$$\begin{aligned} &= 2 \left[x \cdot \int_0^L \sin\left(\frac{n\pi x}{L}\right) - \int_0^L \left[\frac{d}{dx} x \cdot \int_0^L \sin\left(\frac{n\pi x}{L}\right) dx \right] dx \right] \\ &= 2 \left[-\frac{Lx}{n\pi} \cdot \cos\left(\frac{n\pi x}{L}\right) + \int_0^L \frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) dx \right] \\ &= 2 \left[-\frac{Lx}{n\pi} \cdot \cos\left(\frac{n\pi x}{L}\right) + \left(\frac{L}{n\pi}\right)^2 \sin\left(\frac{n\pi x}{L}\right) \right]_0^L \end{aligned}$$

Solving the definite integral we get

$$= (-1)^n \cdot 2 \frac{L^2}{n\pi}$$

Now solving $\frac{2}{L} \int_0^L x^2 \cdot \sin\left(\frac{n\pi x}{L}\right)$

$$\begin{aligned} &= \frac{2}{L} \left[x^2 \cdot \int_0^L \sin\left(\frac{n\pi x}{L}\right) - \int_0^L \left[\frac{d}{dx} x^2 \cdot \int_0^L \sin\left(\frac{n\pi x}{L}\right) dx \right] dx \right] \\ &= \frac{2}{L} \left[-\frac{Lx^2}{n\pi} \cdot \cos\left(\frac{n\pi x}{L}\right) + \int_0^L x \cdot \frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) dx \right] \end{aligned}$$

Applying integration by parts again to evaluate $\int_0^L x \cdot \frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) dx$

$$\begin{aligned} &= \frac{2}{L} \left[-\frac{Lx^2}{n\pi} \cdot \cos\left(\frac{n\pi x}{L}\right) + \left(x \cdot \int_0^L \frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) - \int_0^L \left[\frac{d}{dx} x \cdot \int_0^L \frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) dx \right] dx \right) \right] \\ &= \frac{2}{L} \left[-\frac{Lx^2}{n\pi} \cdot \cos\left(\frac{n\pi x}{L}\right) + \left(x \left(\frac{L}{n\pi} \right)^2 \sin\left(\frac{n\pi x}{L}\right) - \int_0^L \left(\frac{L}{n\pi} \right)^2 \sin\left(\frac{n\pi x}{L}\right) dx \right) \right] \\ &= \frac{2}{L} \left[-\frac{Lx^2}{n\pi} \cdot \cos\left(\frac{n\pi x}{L}\right) + x \left(\frac{L}{n\pi} \right)^2 \sin\left(\frac{n\pi x}{L}\right) + \left(\frac{L}{n\pi} \right)^3 \cos\left(\frac{n\pi x}{L}\right) \right]_0^L \end{aligned}$$

Solving the definite integral we get

$$\begin{aligned} &= \frac{2}{L} \cdot \frac{L^3}{n\pi} \\ &= (-1)^n \cdot 2 \frac{L^2}{n\pi} \end{aligned}$$

Putting the two solutions together

$$B_n = 4 \frac{L^2}{n\pi}$$

Therefore we have deciphered $w(x, t)$ analytically to be

$$w(x, t) = \sum_{n=1}^{\infty} 4 \frac{L^2}{n\pi} \cdot e^{-\alpha \frac{n^2 \pi^2}{L^2} t} \sin\left(\frac{n\pi x}{L}\right)$$

3.2.7 Determining the final solution:

Therefore the final solution is:

$$u(x, t) = u_s + w(x, t)$$

$$u(x, t) = \frac{S_0}{\alpha} \left(\frac{L}{\pi}\right)^2 \sin\left(\frac{\pi x}{L}\right) + \sum_{n=1}^{\infty} 4 \frac{L^2}{n\pi} \cdot e^{-\alpha \frac{n^2 \pi^2}{L^2} t} \sin\left(\frac{n\pi x}{L}\right)$$

3.3 Laplace transform analytical method:

The Laplace transform method is a technique used to simplify differential equations, making them easier to solve. Instead of dealing with both time and space variables at once, we transform the equation so that time-dependent terms become algebraic expressions involving a new variable s called the Laplace parameter. This transformation turns the original partial differential equation into an ordinary differential equation, which is much simpler to solve. Once we solve the equation in this new form, we use the inverse Laplace transform to convert our solution back to the original time-dependent form.

3.3.1 Applying Laplace transform to solve the homogeneous differential equation to find the transient solution

$$\frac{\partial w}{\partial t} = \alpha \frac{\partial^2 w}{\partial x^2}$$

Apply the Laplace transform on both sides we get

$$L\left\{\frac{\partial w}{\partial t}\right\} = L\left\{\alpha \frac{\partial^2 w}{\partial x^2}\right\}$$

Using the Laplace derivative property where for some function a $L\left\{\frac{\partial a}{\partial t}\right\} = sA(x, s) - a(x, 0)$

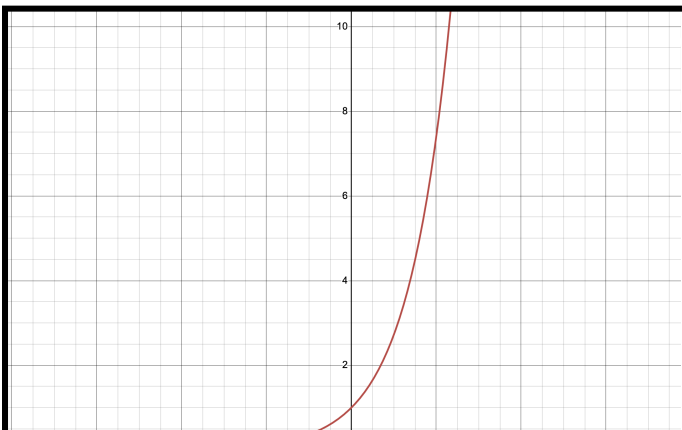
It should be noted that when Laplace transform is applied to both sides $\frac{\partial^2 w}{\partial x^2}$ factor does not get affected since that is the derivative with respect to x whereas the transform function only reduces the time dependent derivatives to algebraic functions. Therefore:

$$sW(x, s) - w(x, 0) = \alpha \frac{\partial^2 W}{\partial x^2}$$

Rearranging by dividing by α :

$$\frac{\partial^2 W}{\partial x^2} - \frac{s}{\alpha} W = -\frac{w(x, 0)}{\alpha}$$

Thus giving us a non-homogeneous ordinary differential equation that is much easier to solve. The Laplace transform method thereby has not eliminated the need to make the assumption we did in **Section 3.2.2** of the previous method and we can solve this further first by finding the homogeneous solution and then the specific solution:



$$\frac{\partial^2 W}{\partial x^2} - \frac{s}{\alpha} W = 0$$

Using the method of trial solution

Let's assume that a particular solution in the form (justified in **Section 3.2.4**)

Graph 3.0 depicting asymptotic nature of e^x

$$W = e^{kx}$$

Substitute W in homogeneous differential equation to find the value of k that satisfies the chosen form of solution:

$$k^2 e^{kx} - \frac{s}{\alpha} e^{kx} = 0$$

Dividing by e^{kx} since $e^{kx} \neq 0$ as evident by the asymptotic nature of the graph depicted in Figure _ where the graph never meets the line $y = 0$.

$$k^2 - \frac{s}{\alpha} = 0$$

Rearranging gives

$$k^2 = \frac{s}{\alpha}$$

$$k = \pm \sqrt{\frac{s}{\alpha}}$$

Therefore the general solution is

$$W_g(x, s) = A e^{\sqrt{\frac{s}{\alpha}}x} + B e^{-\sqrt{\frac{s}{\alpha}}x}$$

3.3.2 Applying the boundary conditions to determine both constants

1) For

$$w(0, t) = W(0, s) = 0$$

Substituting this we get

$$0 = A e^0 + B e^0$$

$$A = -B$$

2) For

$$w(L, t) = W(L, s) = 0$$

Substituting we get

$$0 = A e^{\sqrt{\frac{s}{\alpha}}L} + B e^{-\sqrt{\frac{s}{\alpha}}L}$$

Using $A = -B$ we get

$$0 = A e^{\sqrt{\frac{s}{\alpha}}L} - A e^{-\sqrt{\frac{s}{\alpha}}L}$$

Factoring out A

$$0 = A (e^{\sqrt{\frac{s}{\alpha}}L} - e^{-\sqrt{\frac{s}{\alpha}}L})$$

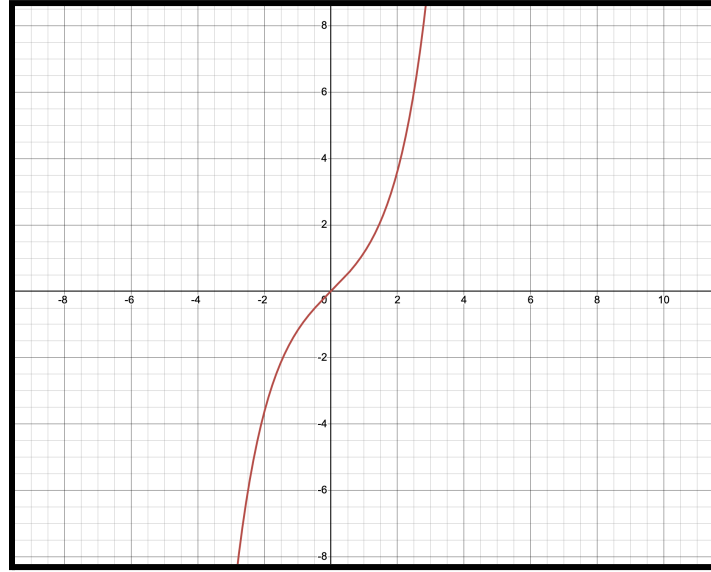
Since

$$e^{\sqrt{\frac{s}{\alpha}}L} - e^{-\sqrt{\frac{s}{\alpha}}L} = 2 \sinh(\sqrt{\frac{s}{\alpha}}L)$$

We get

$$0 = 2A \sinh(\sqrt{\frac{s}{\alpha}}x)$$

Since this hyperbolic function is 0 only when $x = 0$ upon satisfying the above equation as evident from Figure __ it would imply that the equation will only hold true for all values of x when $A = 0$ the general solution $W_g(x, s) = 0$ and disappears



Graph 4.0 depicts the nature of $\sinh(x)$ function

We now solve for the particular solution $W_p(x, s)$ in:

$$\frac{\partial^2 W}{\partial x^2} - \frac{s}{\alpha} W = -\frac{w(x,0)}{\alpha}$$

Where

$$w(x, 0) = f(x) = x(L - x)$$

$$\frac{\partial^2 W}{\partial x^2} - \frac{s}{\alpha} W = -\frac{f(x)}{\alpha}$$

3.3.3 Finding the particular solution for the following term

$$-\frac{f(x)}{\alpha}$$

Since the sine function satisfies the fixed boundary conditions we can approach determining the particular solution using the Fourier sine expansion series as our initial condition $u(x, 0)$ can be decomposed into a sum of sine functions.

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

B_n can be determined using orthogonality as done in **Section 3.2.6** to be

$$B_n = \frac{2}{L} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right)$$

Now we substitute $f(x)$ as the fourier series into the Laplace transform differential equation

$$\frac{\partial^2 W}{\partial x^2} - \frac{s}{\alpha} W = -\frac{1}{\alpha} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

Since sine functions have an orthogonal basis implying that any periodic function can be represented as a sum of these sine and cosine functions, and the integral of the product of two different sine functions (or a sine and cosine function) over a period is zero we can assume that $W(x, s)$ will have a sine series form:

$$W(x, s) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

Taking the second derivative gives

$$\frac{\partial^2 W}{\partial x^2} = \sum_{n=1}^{\infty} A_n - \left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right)$$

Substituting this back into the transformed differential equation gives

$$\sum_{n=1}^{\infty} A_n - \left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right) - \frac{s}{\alpha} \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) = -\frac{1}{\alpha} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

Taking $\sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$ common

$$\sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) \left[-\left(\frac{n\pi}{L}\right)^2 - \frac{s}{\alpha} \right] = -\frac{1}{\alpha} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

Comparing coefficients of $\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)$ we get

$$A_n \left[-\left(\frac{n\pi}{L}\right)^2 - \frac{s}{\alpha} \right] = -\frac{1}{\alpha} B_n$$

Simplifying gives the following:

$$A_n \left[\left(\frac{n\pi}{L}\right)^2 + \frac{s}{\alpha} \right] = \frac{1}{\alpha} B_n$$

$$A_n = \frac{B_n}{\alpha \left[\left(\frac{n\pi}{L}\right)^2 + \frac{s}{\alpha} \right]}$$

$$A_n = \frac{B_n}{[\alpha \left(\frac{n\pi}{L}\right)^2 + s]}$$

Substituting A_n into our assumed function for $W(x, s)$

$$W(x, s) = \sum_{n=1}^{\infty} \frac{B_n}{[\alpha \left(\frac{n\pi}{L}\right)^2 + s]} \sin\left(\frac{n\pi x}{L}\right)$$

3.3.4 Finding the original function with time dependency

Now that we have determined the Laplace transform solution we need to find its inverse to find out original function and time domain $u(x, t)$

We will consider the inverse Laplace formula for the function of the form

$$L^{-1}\left(\frac{1}{s+a}\right) = e^{-at}$$

We can compare this to the denominator in our expression by writing

$$[\alpha \left(\frac{n\pi}{L}\right)^2 + s] \text{ as } [s + \alpha \left(\frac{n\pi}{L}\right)^2]$$

Using the inverse Laplace formula above we get

$$L^{-1}\left(\frac{1}{s + \alpha \left(\frac{n\pi}{L}\right)^2}\right) = e^{-\alpha \left(\frac{n\pi}{L}\right)^2 t}$$

Now we can take the inverse Laplace of $W(x, s)$

$$L^{-1}(W(x, s)) = \sum_{n=1}^{\infty} B_n e^{-\alpha(\frac{n\pi}{L})^2 t} \sin(\frac{n\pi x}{L})$$

Therefore:

$$w(x, t) = \sum_{n=1}^{\infty} B_n e^{-\alpha(\frac{n\pi}{L})^2 t} \sin(\frac{n\pi x}{L})$$

3.3.5 Finding the steady state solution

Now we will determine the steady state solution u_s using the Laplace transform method

Steady state solution means the temperature does not change with time and $\frac{\partial u}{\partial t} = 0$. Therefore the PDE becomes

$$\alpha \frac{\partial^2 u_s}{\partial x^2} + S_0 \sin(\frac{\pi x}{L}) = 0$$

Dividing by α

$$\frac{\partial^2 u_s}{\partial x^2} = -\frac{S_0}{\alpha} \sin(\frac{\pi x}{L})$$

We apply Laplace transform to $u_s(x)$

$$L\{u_s(x)\} = U_s(p)$$

Using the Laplace property for second derivatives we get

$$L\{u_s(x)\} = U_s(p) = p^2 U_s(p) - p u_s(0) - u'_s(0)$$

Taking the Laplace transform on both sides gives

$$p^2 U_s(p) - p u_s(0) - u'_s(0) = -\frac{S_0}{\alpha} L[\sin(\frac{\pi x}{L})]$$

Laplace transform of $\sin(\frac{\pi x}{L})$ using known formula

$$L[\sin(ax)] = \frac{a}{p^2 + a^2}$$

We get

$$L[\sin(\frac{\pi x}{L})] = \frac{\frac{\pi}{L}}{p^2 + (\frac{\pi}{L})^2}$$

Thus the equation becomes

$$p^2 U_s(p) - p u_s(0) - u'_s(0) = -\frac{S_0}{\alpha} \frac{\frac{\pi}{L}}{p^2 + (\frac{\pi}{L})^2}$$

Since we are solving for the steady state we can make the following assumption as we have previously

$$u_s(0) = 0 \text{ and } u'_s(0) = 0$$

Therefore

$$U_s(p) = -\frac{S_0}{\alpha} \frac{\frac{\pi}{L}}{[p^2 + (\frac{\pi}{L})^2] p^2}$$

We can write the above equation as

$$U_s(p) = -\frac{S_0}{\alpha} \cdot \frac{\pi}{L} \cdot \frac{1}{[p^2 + (\frac{\pi}{L})^2] p^2}$$

We can see that the significant denominator of the above function is made up of $p^2 + (\frac{\pi}{L})^2$ and p^2

And we know from the standard inverse Laplace forms that

$$L^{-1}\left[\frac{1}{p^2+a^2}\right] = \frac{\sin(ax)}{a}$$

$$L^{-1}\left[\frac{1}{p^2}\right] = x$$

We can use partial fractions to get the above forms by assuming $p^2 = j$

$$\frac{1}{[j+(\frac{\pi}{L})^2]j} = \frac{A}{j} + \frac{B}{j+(\frac{\pi}{L})^2}$$

Taking LCM on RHS and multiplying with

$$A[j + (\frac{\pi}{L})^2] + B(j) = 1$$

Expanding

$$Aj + Bj + A(\frac{\pi}{L})^2 = 1$$

Comparing coefficients we get

$$A + B = 0 \quad \text{and} \quad A(\frac{\pi}{L})^2 = 1$$

Hence we get

$$A = (\frac{L}{\pi})^2$$

$$B = -(\frac{L}{\pi})^2$$

Therefore we get

$$\frac{1}{[j+(\frac{\pi}{L})^2]j} = \frac{(\frac{L}{\pi})^2}{j} - \frac{(\frac{L}{\pi})^2}{j+(\frac{\pi}{L})^2}$$

Substituting j back as p^2 gives

$$\frac{1}{[p^2+(\frac{\pi}{L})^2]p^2} = \frac{(\frac{L}{\pi})^2}{p^2} - \frac{(\frac{L}{\pi})^2}{p^2+(\frac{\pi}{L})^2}$$

We can now substitute this back into the main transformed differential equation to get

$$U_s(p) = -\frac{S_0}{\alpha} \cdot \frac{\pi}{L} \cdot (\frac{L}{\pi})^2 \cdot (\frac{1}{p^2} - \frac{1}{p^2+(\frac{\pi}{L})^2})$$

$$U_s(p) = -\frac{S_0}{\alpha} \cdot \frac{L}{\pi} \cdot (\frac{1}{p^2} - \frac{1}{p^2+(\frac{\pi}{L})^2})$$

Now we can apply the standard inverse Laplace discussed above to get

$$L^{-1}(\frac{1}{p^2} - \frac{1}{p^2+(\frac{\pi}{L})^2}) = x - \frac{L}{\pi} \sin(\frac{\pi}{L}x)$$

And find $u_s(x)$ to be

$$u_s(x) = -\frac{S_0}{\alpha} \cdot \frac{L}{\pi} \cdot (x - \frac{L}{\pi} \sin(\frac{\pi}{L}x))$$

Opening the bracket we get

$$u_s(x) = -\frac{S_0}{\alpha} \cdot \frac{L}{\pi} x + \frac{S_0}{\alpha} \cdot (\frac{L}{\pi})^2 \sin(\frac{\pi}{L}x)$$

3.3.6 Formulating the final solution

Therefore the solution $u(x, t)$ is given by

$$u(x, t) = u_s(x) + w(x, t)$$

$$u(x, t) = -\frac{S_0}{\alpha} \cdot \frac{L}{\pi} x + \frac{S_0}{\alpha} \cdot \left(\frac{L}{\pi}\right)^2 \sin\left(\frac{\pi}{L}x\right) + \sum_{n=1}^{\infty} B_n e^{-\alpha\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi x}{L}\right)$$

By solving the transformed equation, we obtain a solution that includes an extra linear term in $x - \frac{S_0}{\alpha} \cdot \frac{L}{\pi} x$ compared to the solution from the separation of variables method. This term arises because the Laplace transform method directly accounts for the internal heat source $S_0 \sin(\frac{\pi x}{L})$ as an explicit driving force rather than decomposing it into sinusoidal components. Physically, this means that the Laplace method captures an additional steady-state gradient contribution, which might not be immediately apparent when using a purely Fourier-based approach.

This process demonstrates the power of the Laplace transform in handling non-homogeneous PDEs with forcing terms more effectively than separation of variables, making it a more general and insightful approach to solving heat conduction problems. This has been further evaluated in **Section 4.0**.

3.4 Variation method

Variational method is a method that is used to solve partial differential equations without directly finding the analytical solution and solving the differential equation but instead applying a condition to the PDE that the solution to the PDE must fulfill. This method therefore imposes small variations in and culminates the response of the PDE to this variation to piece together the final solution and distribution of temperature in the 1D rod.

3.4.1 applying the variational method using test function method

To apply the variational method, we first multiply both sides of the PDE by a test function $v(x)$, which belongs to a suitable function space:

$$v(x) \left[\frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} - S_0 \sin\left(\frac{\pi x}{L}\right) \right] = 0$$

Now we must integrate both sides over the domain $0 \leq x \leq L$

$$\int_0^L v(x) \left[\frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} - S_0 \sin\left(\frac{\pi x}{L}\right) \right] dx = 0$$

This integral enforces the equation through a small variation, meaning that the solution function $u(x, t)$ must satisfy the equation overall rather than pointwise.

Now there is a small evident problem here which is the $\frac{\partial^2 u}{\partial x^2}$ second order term because it involves higher-order derivatives.

3.4.2 simplifying the variation integral

To simplify, we use integration by parts to reduce its order.

First we open the bracket and expand the single integral into 3 integrals:

$$\int_0^L v(x) \frac{\partial u}{\partial t} dx - \int_0^L v(x) \alpha \frac{\partial^2 u}{\partial x^2} dx - \int_0^L v(x) S_0 \sin\left(\frac{\pi x}{L}\right) dx = 0$$

Now we will simplify $\int_0^L v(x) \alpha \frac{\partial^2 u}{\partial x^2}$ using integration by parts

For two functions $f(x)$ and $g(x)$ integration by parts is given by

$$\int_0^L f(x)g''(x)dx = f(x)g'(x)|_0^L - \int_0^L f'(x)g'(x)dx$$

Therefore for our case we will consider $f(x) = v(x)$ and $g(x) = u(x, t)$

$$\int_0^L v(x) \alpha \frac{\partial^2 u}{\partial x^2} dx = v(x) \frac{\partial u}{\partial x} \Big|_0^L - \int_0^L \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx$$

Since our chosen boundary conditions enforces $u(0, t) = u(L, t) = 0$ the boundary term $v(x) \frac{\partial u}{\partial x} \Big|_0^L$ vanishes and therefore the equation simplifies to give

$$\int_0^L v(x) \alpha \frac{\partial^2 u}{\partial x^2} dx = - \int_0^L \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx$$

Substituting this back into the original integral we get

$$\int_0^L v(x) \frac{\partial u}{\partial t} dx + \int_0^L \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx = \int_0^L v(x) S_0 \sin\left(\frac{\pi x}{L}\right) dx$$

3.4.3 Simplifying the solution function $u(x,t)$ using eigenfunctions and values

Now we need to simplify our required function $u(x,t)$ and therefore we assume it to be a sum of basic functions called eigenfunctions.

In order to understand the significance of this step we must first understand what a laplacian term is. In the heat equation $\frac{\partial^2 u}{\partial x^2}$ the term is called the laplacian. Physically, the Laplacian measures how the function $u(x,t)$ differs from its immediate surroundings, which in turn determines how heat diffuses over time.

If a function is an eigenfunction of the Laplacian, it means that applying the Laplacian to the function just scales it rather than changing its shape such that

$$\frac{\partial^2 \phi_n}{\partial x^2} = -\lambda_n \phi_n$$

where λ is a constant called the eigenvalue. This is important because it allows us to decompose any function into these fundamental building blocks and solve for each component separately.

Eigenfunctions of the Laplacian form a natural basis for solving differential equations with boundary conditions. The key reason is orthogonality, which helps us decouple the PDE into simpler problems for each eigenfunction.

This can be compared to simplifying a complicated sound wave into simpler sine waves as in Fourier series. Similarly in this case we express $u(x,t)$ in terms of simpler functions and their multiples (given by the eigenvalue) that satisfy the boundary conditions.

Since the heat equation involves the Laplacian, and we know its eigenfunctions behave predictably, it makes sense to assume $u(x, t)$ can be written as a sum of these eigenfunctions:

$$u(x, t) = \sum_{n=1}^{\infty} c_n(t) \phi_n(x)$$

Where $\phi_n(x)$ is the eigenfunction or functions of the laplacian that will be determined next and $c_n(t)$ is a time dependent coefficient that must be added when solving partial differential equations to ensure time dependency of the function.

The reason eigenfunction/functions is stated above is because the function is a family of curves based on the value of n , however while solving the differential equation we consider the smallest value of n to find the fundamental equation and avoid multiplicity with big coefficients.

In order to determine the eigenfunction and eigenvalue we will solve the eigenvalue problem for the laplacian

$$\frac{\partial^2 \phi_n}{\partial x^2} = -\lambda_n \phi_n \text{ where } \phi_n(0) = 0 \text{ and } \phi_n(L) = 0$$

Since this is a second order homogeneous differential equation we can solve this using the method of general solution that we used in **Section 3.2**. We will assume that the solution is in the general form as done in **Section 3.2.3** therefore getting:

$$\phi_n = A \cos(\sqrt{\lambda_n} x) + B \sin(\sqrt{\lambda_n} x)$$

The above form was chosen so that the form of the equation obtained will be similar to the previous answers we have received thus making them comparable.

3.4.4 Applying boundary conditions

Now we can apply the boundary conditions as we did earlier in **Section 3.2.3** as well

For when $\phi_n(0) = 0$

$$0 = A \cos(0) + B \sin(0) \\ A = 0$$

For when $\phi_n(L) = 0$

$$0 = B \sin(L\sqrt{\lambda})$$

For a non-trivial solution where $B \neq 0$ we need

$$\sin(L\sqrt{\lambda}) = 0$$

to satisfy the equation.

This is only possible when

$$L\sqrt{\lambda} = n\pi \text{ where } n = 1, 2, 3...$$

So,

$$\lambda_n = \frac{n^2 \pi^2}{L^2}$$

Hence

$$\phi_n(x) = B \sin\left(\frac{n\pi x}{L}\right)$$

Now that we have determined the eigenfunction we can write $u(x, t)$ as

$$u(x, t) = \sum_{n=1}^{\infty} c_n(t) B \sin\left(\frac{n\pi x}{L}\right)$$

Through this approach we were able to transform a partial differential equation into a system to ordinary differential equations thus making it easier to solve.

3.4.5 Reformulating the initial variation integral from Section 3.4.1

Now we must substitute this back into the original variation integral this getting

$$\int_0^L v(x) \frac{\partial u}{\partial t} dx + \int_0^L \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx = \int_0^L v(x) S_0 \sin\left(\frac{\pi x}{L}\right) dx$$

3.4.6 Solving the variation integral

In order to solve this integral we must first make the following assumption

$$v(x) = \sin\left(\frac{m\pi x}{L}\right)$$

The above assumption was made because it has a similar nature to the obtained eigenfunction of the Laplacian under Dirichlet boundary conditions, meaning it naturally satisfies $v(0) = v(L) = 0$. This choice simplifies the problem by converting the partial differential equation into an algebraic system when applying the weighted residual method. Additionally, sine functions form an orthogonal basis, making it easier to extract coefficients and ensure convergence of the solution. Since the heat equation inherently involves the Laplacian, expanding the solution in terms of the obtained eigenfunctions ensures an accurate representation of the transient behavior of the system.

Now substituting the obtained $u(x, t)$ and assumed $v(x)$ into the variation integral we get

$$\begin{aligned} \sum_{n=1}^{\infty} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \frac{dc_n(t)}{dt} B \sin\left(\frac{n\pi x}{L}\right) dx + \alpha \sum_{n=1}^{\infty} \int_0^L c_n(t) \frac{d}{dx} \sin\left(\frac{m\pi x}{L}\right) \frac{d}{dx} \sin\left(\frac{n\pi x}{L}\right) dx &= \int_0^L \sin\left(\frac{m\pi x}{L}\right) S_0 \sin\left(\frac{\pi x}{L}\right) dx \\ \int_0^L \sin\left(\frac{m\pi x}{L}\right) \frac{dc_n(t)}{dt} B \sin\left(\frac{n\pi x}{L}\right) dx + \alpha c_n(t) \int_0^L \frac{m\pi}{L} \sin\left(\frac{m\pi x}{L}\right) \cdot \frac{n\pi}{L} \sin\left(\frac{n\pi x}{L}\right) dx &= \int_0^L \sin\left(\frac{m\pi x}{L}\right) S_0 \sin\left(\frac{\pi x}{L}\right) dx \end{aligned}$$

Now we will simplify $\int_0^L \frac{m\pi}{L} \sin\left(\frac{m\pi x}{L}\right) \cdot \frac{n\pi}{L} \sin\left(\frac{n\pi x}{L}\right) dx$ using orthogonality

Taking constants out and simplifying we get

$$\frac{m\pi}{L} \cdot \frac{n\pi}{L} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

We have already in **Section 3.2.6** determined the solution to the integral when $n = m$ and therefore solving the above we get:

$$\frac{m\pi}{L} \cdot \frac{n\pi}{L} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \cdot \sin\left(\frac{n\pi x}{L}\right) dx = \frac{m^2 \pi^2}{2L}$$

Substituting this simplification gives us:

$$\frac{dc_m(t)}{dt} \frac{L}{2} + \alpha c_n(t) \frac{m^2 \pi^2}{2L} = S_0 \int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{\pi x}{L}\right) dx$$

Consider $m = 1$ we will evaluate $\int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{\pi x}{L}\right) dx$

$$\int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{L}\right) dx$$

Which also becomes $\frac{L}{2}$ as discussed in previous solutions

Therefore we get

$$\frac{dc_1(t)}{dt} \frac{L}{2} + \alpha c_1(t) \frac{\pi^2}{2L} = \frac{S_0 L}{2}$$

Simplifying we get

$$\frac{dc_1(t)}{dt} + \alpha c_1(t) \frac{\pi^2}{L^2} = S_0$$

3.4.7 Determining $c_1(t)$ using integrating factor

Now we can solve the above differential equation to determine $c_1(t)$ using the integrating factor method

Where the integration factor

$$IF = e^{\int \alpha \frac{\pi^2}{L^2} dt}$$

$$IF = e^{\alpha \frac{\pi^2}{L^2} t}$$

Substituting IF into the integrating factor formula given by

$$IF \cdot y = \int IF \cdot q(x) dx$$

Considering $q(x) = S_0$ and $y = c_1(t)$

$$c_1(t) e^{\alpha \frac{\pi^2}{L^2} t} = S_0 \int e^{\alpha \frac{\pi^2}{L^2} t} dt$$

Therefore

$$c_1(t) e^{\alpha \frac{\pi^2}{L^2} t} = S_0 e^{\alpha \frac{\pi^2}{L^2} t} + C$$

Dividing by $e^{\alpha \frac{\pi^2}{L^2} t}$

$$c_1(t) = S_0 + C e^{-\alpha \frac{\pi^2}{L^2} t}$$

3.4.8 Determining integration constant C

Now we will use the initial condition $u(x, 0) = x(L - x)$ to determine the constant C

For $u(x, 0) = c_1(0) = x(L - x)$

Therefore taking $t = 0$ in $u(x, t) = \sum_{n=1}^{\infty} c_n(t) B \sin\left(\frac{n\pi x}{L}\right)$ we get

$$u(x, 0) = \sum_{n=1}^{\infty} c_n(0) B \sin\left(\frac{n\pi x}{L}\right)$$

$$x(L - x) = \sum_{n=1}^{\infty} c_n(0) B \sin\left(\frac{n\pi x}{L}\right)$$

Multiplying both sides by $\sin\left(\frac{n\pi x}{L}\right)$ and taking integral for $0 \leq x \leq L$ to apply orthogonality as we have previously we get

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) x(L - x) dx = \int_0^L c_n(0) B \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

Since $c_n(0)$ is a function of t we can take it out of the integral

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) x(L - x) dx = c_n(0) \int_0^L B \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

We have already solved the integrals on both sides so substituting from **Section 3.2.6** thus getting

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) x(L - x) dx = \frac{BL}{2} c_n(0)$$

$$\frac{2}{BL} \int_0^L \sin\left(\frac{n\pi x}{L}\right) x(L - x) dx = c_n(0)$$

$$\frac{2}{BL} \cdot 4 \frac{L^2}{n\pi} = c_n(0)$$

$$c_n(0) = 8 \frac{L}{Bn\pi}$$

Substituting this into the equation for $c_1(t)$ we get

$$c_1(0) = S_0 + C e^{-\alpha \frac{\pi^2}{L^2} 0}$$

$$C = S_0 - c_1(0)$$

Substituting $c_n(0) = 8 \frac{L}{Bn\pi}$ for $n=1$

$$C = S_0 - 8 \frac{L}{B\pi}$$

Therefore

$$c_1(t) = S_0 + (S_0 - 8 \frac{L}{B\pi}) e^{-\alpha \frac{\pi^2}{L^2} t}$$

3.4.9 Culminating the final solution

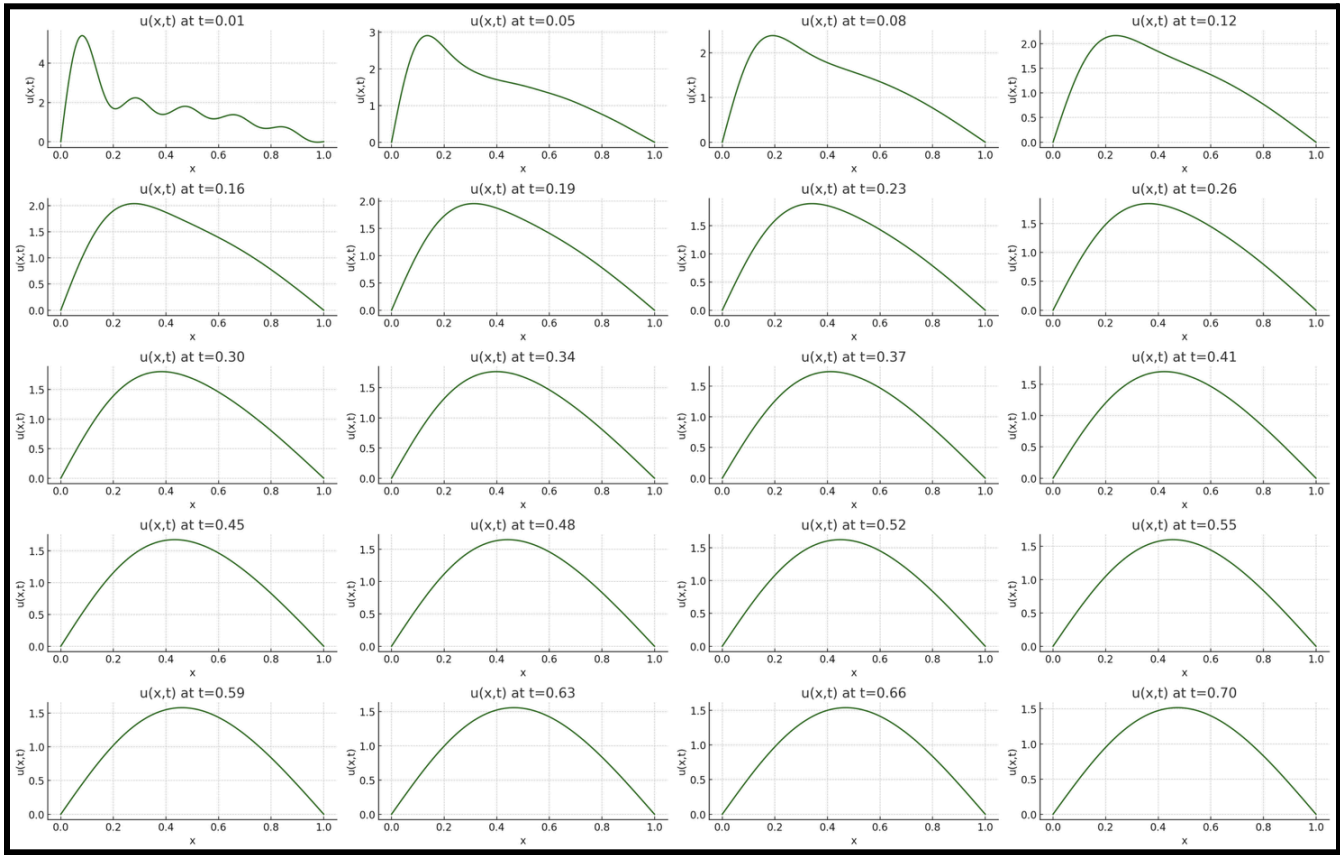
Therefore our final solution for one family of curves $n = 1$ is

$$u(x, t) = (S_0 + (S_0 - 8\frac{L}{B\pi})e^{-\alpha\frac{\pi^2}{L^2}t}) \sum_{n=1}^{\infty} B\sin(\frac{n\pi x}{L})$$

The above solution matches that found by the previous two methods however based on the assumptions made we were not able to find the general solution for $(S_0 + (S_0 - 8\frac{L}{B\pi})e^{-\alpha\frac{\pi^2}{L^2}t})$ term of the solution in terms of n whereas the laplace transform function was able to do so effectively.

4.0 Visualization of the final solution

In order to demonstrate the solutions found I wrote a code in python to produce more graphs more detailed than a 2D line graph by incorporating 3D visualizations and colour optimization to show the temperature gradient.



Graph 5.0 depicts the line graph of the evolution of the solution $u(x,t)$ with x .

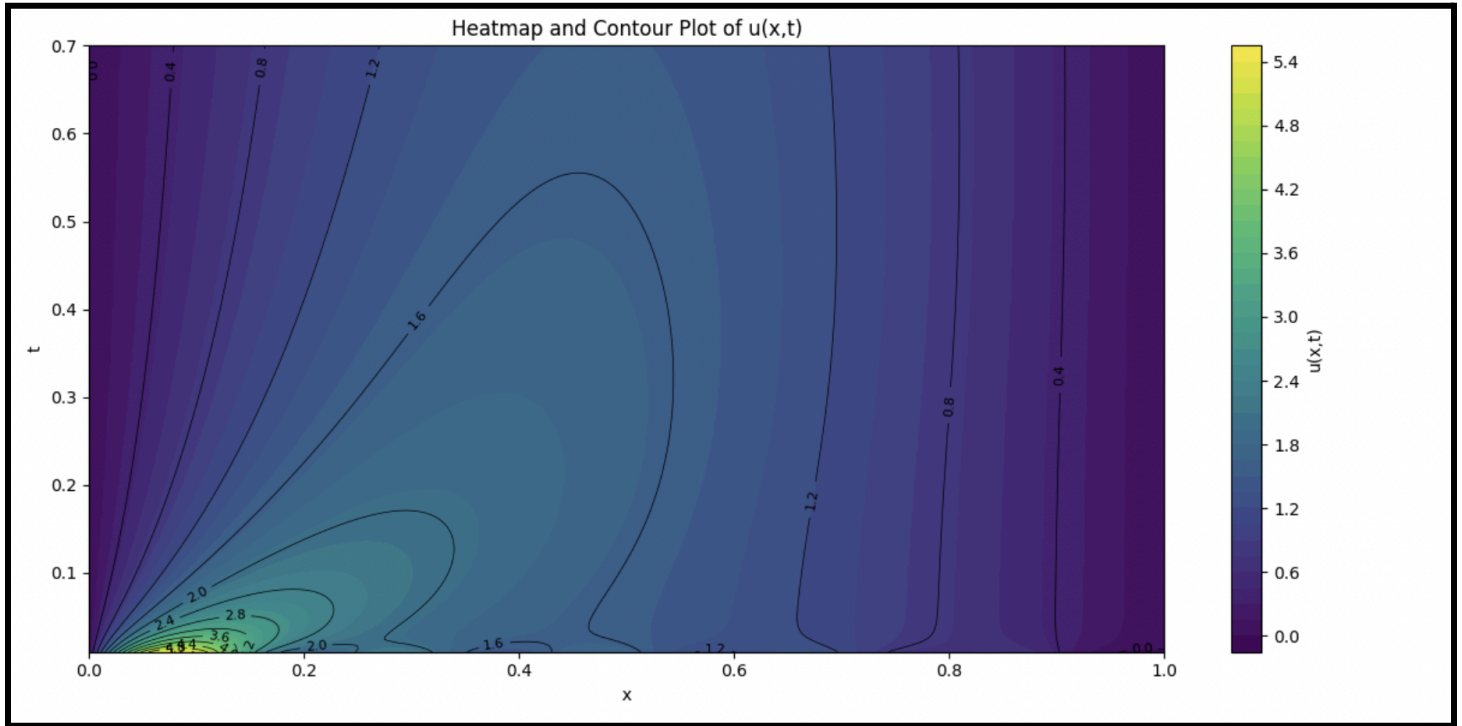
Mathematical Interpretation:

The line graphs illustrate the time dependent evolution of the function $u(x, t)$, where each individual plot represents the solution at a specific time t . This solution arises from a Fourier series that includes both a steady state term and a sum of

transient terms that decay exponentially over time due to the presence of $e^{-\alpha\frac{\pi^2}{L^2}t}$. Plots where $t < 0.5$ show complex behavior influenced by higher order sine terms, but as t increases, these terms diminish rapidly, leaving behind the fundamental solution. This visualizes the natural damping mechanism inherent in the system's partial differential equation.

Physical Interpretation:

Physically the graphs represent how temperature changes along the spatial domain over time. The system initially exhibits sharp oscillations due to high frequency components from the chosen non-equilibrium initial condition due to higher modes. As time progresses these oscillations fade depicting the dissipation of energy and the transition toward a stable equilibrium state. The shape gradually simplifies into a smooth sine graph depicting how the physical system is governed by diffusion and damping effects.



Graph 6.0 depicts a contour plot to show the heat distribution in a one dimensional rod given by $u(x,t)$

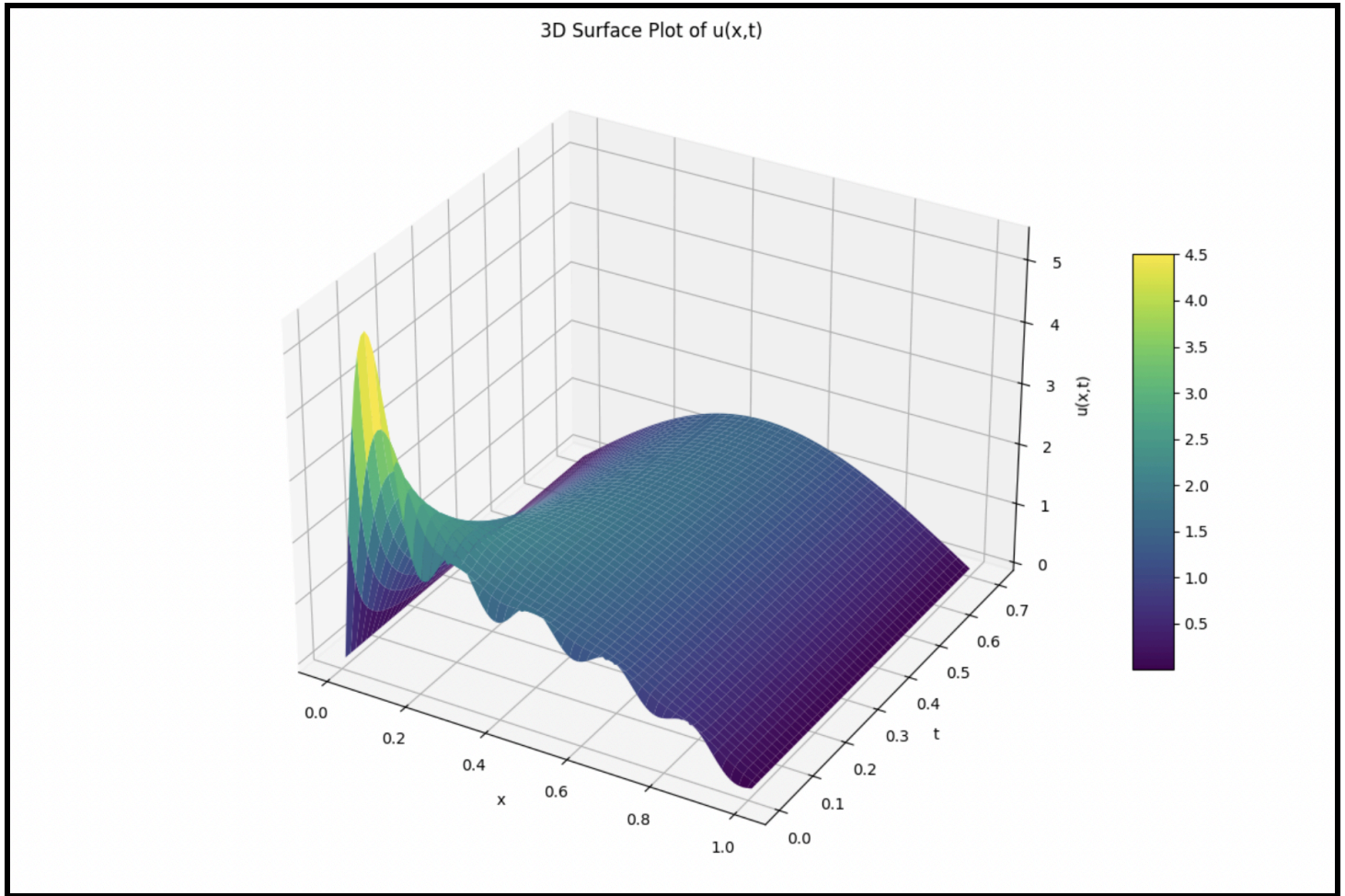
Mathematical Interpretation:

The 3D surface plot offers a continuous and unified representation of $u(x, t)$ over space and time. The vertical axis shows the magnitude of the solution, while the horizontal axes cover space (x) and time (t). From a mathematical perspective, this visualization reveals how the solution transitions from a high-frequency, composite waveform to a low-frequency,

stable sine function. The decay of each harmonic term, governed by the exponential factor $e^{-\alpha \frac{\pi^2}{L^2} t}$, is visible as surface ripples flatten over time. The 3D view allows us to simultaneously observe the convergence behavior of the Fourier series and the effect of time on each spatial point.

Physical Interpretation:

Physically, this plot is analogous to watching the evolution of the heated rod over time. Peaks and valleys near the front of the graph (when $t < 2.0$) indicate localized disturbances and rapid variations. As you move along the time axis, these features smooth out, indicating diffusion or damping processes at work. It encapsulates the behavior of the system, making it clear how energy spreads or dissipates.



Graph 7.0 depicts the 3D plot of the function to show both time and spatial dependency of the function $u(x,t)$

Mathematical Interpretation:

The heatmap with contour lines condenses the solution $u(x, t)$ into a two dimensional plot where color intensity represents the function's magnitude of temperature. This graph is especially useful as it retains both spatial and temporal dependencies and highlights gradient behavior through color transitions and contour intervals. It efficiently displays the fading behaviour of the transient components in the Fourier series with color gradients tracking how each mode's influence fades over time. The contour lines drawn at equal values of $u(x, t)$, act like level sets, offering insight into how fast or slowly the function is changing with respect to time and space.

Physical Interpretation:

This graph is similar to viewing a thermal scan of a material, where color indicates temperature. The plot allows for a visual assessment of where changes are most rapid in areas with tightly packed contours and sharp color shifts and where the system is stable. The upper region ((when $t < 2.0$) displays a more complex and varied distribution, while the lower region ((when $t > 2.0$) shows a more uniform pattern, consistent with diffusion or damping effects in the physical system.

5.0 Comparison of methods

The process of solving the heat equation through multiple approaches including separation of variables, Laplace transforms, and Variation method allowed for a rigorous evaluation of the strengths and limitations of each technique. Each method provided unique insights into the behavior of the heat equation, yet also exposed computational or theoretical constraints, which were subsequently refined by the next approach satisfying the aim of this exploration which is understanding mathematical thinking and the invention of newer approaches to solving the same question in the context of the 1D heat equation.

5.1 Separation of Variables:

The separation of variables method offered an intuitive analytical solution by decomposing the heat equation into steady-state and transient components. This method was advantageous because its structured framework allowed the solutions to be easily understood through existing methods thereby expressing its expansions as sine series. However, it imposed significant constraints on boundary conditions, requiring homogeneous conditions to be effective. The method also struggled with handling non-homogeneous forcing terms like the additional sinusoidal source term in my equation. Furthermore, while it provided explicit forms for transient and steady-state solutions, getting to the solution entailed a lot of mathematical steps and was very time consuming.

5.2 Laplace Transform:

The Laplace transform method addressed some of the limitations of separation of variables by allowing for non-homogeneous boundary conditions and initial conditions to be incorporated more naturally through the methodology itself. The transformation converted the PDE into an ODE in the Laplace domain which was easily manipulated to then be inverted thus yielding the expected general solution. Furthermore this method introduced an additional factor in the final solution as seen in the equation obtained that demonstrated how the separation of variables method had ignored certain transient effects, which were captured by the Laplace approach. However, the challenge of inverting the Laplace transform for complex boundary conditions and the lack of physical intuition in the transformed space made this method less accessible for direct interpretation and solving the inverse Laplace transform analytically became more intensive, requiring something known as contour integration or approximation techniques in many cases which was beyond the scope of this exploration.

5.3 Variation method :

The variational method reformulated the heat equation by optimizing it rather than solving the PDE directly. By employing the calculus of variations that entails making small changes in the function to find maximum and minimum points helps find solutions by reducing just these peaks thus providing deeper insight into heat distribution and equilibrium. Unlike separation of variables, which imposes restrictive boundary conditions, or Laplace transforms which require complex inversions, the variational method naturally incorporates necessary conditions and the non-homogeneous term to solve the PDE thus making it particularly effective for problems where traditional methods struggle. However, the method introduced many challenges and was the most complex to learn.

5.4 Why the Variation method was considered even though Laplace transform gave a more precise answer:

The variational method stands out because it provides a way to solve PDEs by identifying significant changes in the function with respect to the two dependent variables rather than explicitly solving the differential equation. This offered a fundamentally different perspective on the problem, emphasizing on physical implication of the function rather than

finding the explicit functional form of solution. Unlike separation of variables, which struggles with arbitrary forcing functions, and Laplace transforms, which require inversion techniques, the variational approach directly yielded an approximate solution that could further be systematically refined however requiring large computational power but proving to be accurate for the context chosen.

6.0 Conclusions and Implications:

This comparative study therefore highlights the evolution of mathematical techniques in solving partial differential equations particularly in heat transfer applications. Each method builds upon the limitations of the previous, reflecting the broader mathematical principle that no single method is universally optimal but instead different approaches serve different problem constraints. The Variation method stands out as the most powerful and adaptable method for the given scenario. It not only maintains the theoretical advantage of analytical solutions but also introduces a computationally feasible alternative for real-world applications. This method can be extended to higher-dimensional heat equations unlike the Laplace transform method that would become exponentially complex upon application to higher dimensions.

7.0 Limitations:

7.1. Assumptions on the Nature of the Heat Equation

A fundamental limitation of this study is the assumption that the heat equation remains linear with constant thermal diffusivity α . In real world applications, heat conduction often involves nonlinear effects like temperature dependent material properties, phase changes, or heat loss through convection currents.

Implication: The chosen solution methods, particularly separation of variables and the variational method, assume linear superposition, which does not hold in nonlinear PDEs. This limits the direct applicability of these techniques to real-world thermal systems where the thermal conductivity constant varies with temperature.

Improvement: To extend the applicability of this approach, perturbation methods or numerical iterative solvers such as finite element or finite difference methods could be integrated with the variational method to handle nonlinearity but would require large computational power to derive numerical steps required to find the numerical solution.

7.2. Constraints of the Variational Method

- **Choice of Test Functions:** The accuracy of the solution heavily depends on the selection of appropriate test functions. Poor choices could lead to inaccurate or slow-converging results as in our case where we only found the convergence at $n = 1$ and could not find a general solution for a part of the desired equation using the variational method.
- **Computational Intensity:** The method often requires solving complex integrals, especially when applying integration by parts several times or dealing with higher-order differential operators making it a lengthy process.
- **Boundary Condition Sensitivity:** Incorporating complex boundary conditions required careful formulation of function spaces and constraints, which complicated the derivation making it hard to construct the final solution.
- **Numerical Approximation Issues:** In practical scenarios, analytical solutions are rarely feasible, and numerical techniques like the finite element method are only feasible as analytical solutions introduce discretization errors. This study could not however focus on numerical solutions due to unavailability of the required computing power.

Improvement: To address these challenges, adaptive function space selection techniques, such as orthogonal polynomials or wavelets, can enhance convergence. Numerical techniques like Galerkin's method can also be optimized using

automated basis selection. Additionally machine learning can be used for variational function approximation which could further allow for application of numerical approximations.

7.3. Laplace Transform Inversion Challenges

The Laplace transform method provided a more general solution but introduced difficulties in inverting the transform, particularly when the inverse formula led to complex contour integrals. This limitation becomes significant when handling real-world boundary conditions that do not allow for straightforward inverse Laplace transformations.

Improvement: A possible enhancement would be to numerically invert the Laplace transform using techniques like:

- Talbot's Method for better numerical inversion for more complex problems
- Post-Widder Method for time domain reconstruction that would eliminate the lengthy steps for solving the transient time dependent solution like we did in **Section 3.3**.
- Fractional derivatives based Laplace inversion, which is useful in heat transfer with memory effects but is beyond the scope of this study.

Researching and finding these high order methods allowed the understanding of the vastness of mathematical tools and models and how every problem encountered throughout the exploration had a plausible solution.

7.4. Extension to Multi-Dimensional Cases

This study was constrained to a 1D heat conduction problem, which, while insightful, does not fully represent the complexity of real-world heat transfer problems. In practical applications:

- Heat conduction occurs in 2D and 3D, introducing additional spatial complexity.
- Convection and radiation effects further complicate heat transfer models.

Improvement: The methods developed in this study could be extended to higher dimensions by but are at this level beyond the scope of this exploration:

- Using the variational method for 2D/3D PDE however computational complexity is increased significantly.
- Adapting interpolation techniques to multi-dimensional grids would however require knowledge about spline interpolation as that would be most fitting for this problem.
- Exploring finite element based solutions to approximate the response in complex geometries however numerical approaches were beyond the scope of this exploration.

7.5. Limited Consideration of Numerical Stability and Convergence

Although the mathematical rigor of each method was analyzed, the study did not extensively explore numerical stability and convergence. Some issues that might arise include:

- Fourier series truncation errors (in separation of variables) which is an error of approximation made when an infinite element is considered finite.
- Numerical instability in inverse Laplace transforms.
- Discretization errors in numerical variational solutions where discontinuities or jumps in the strain between the relationship between two variables occur due to misrepresentation of the stress applied by the boundary conditions on the function.

Improvement: Future studies should include quantitative error analysis, such as:

- Comparing analytical and numerical solutions using error metrics like Mean Squared Error.
- Evaluating convergence criteria for each method.
- Sensitivity analysis to determine how small changes in initial/ boundary conditions impact final results.

8.0 Extension:

The study can be extended to 2D or 3D heat conduction problems, where the Variation method becomes a significant advantage. The interpolation technique could then be modified using higher-dimensional basis functions, such as bilinear or bicubic interpolation in 2D and trivariate interpolation in 3D. Furthermore the recent rise in data driven physics based models, training a neural network to learn the Variations methods, test fuction's response for a given heat equation and then applying interpolation techniques to approximate solutions more efficiently would be interesting. This would bridge classical PDE solving techniques with modern computational methods, opening doors to AI-assisted engineering simulations.

Bibliography

Books

Cannon, John R. *The One-Dimensional Heat Equation*. Addison-Wesley, 1984.

Crank, John. *The Mathematics of Diffusion*. 2nd ed., Oxford University Press, 1975.

Evans, Lawrence C. *Partial Differential Equations*. 2nd ed., American Mathematical Society, 2010.

Logan, J. David. *Applied Partial Differential Equations*. 3rd ed., Springer, 2015.

Renardy, Michael, and Robert C. Rogers. *An Introduction to Partial Differential Equations*. 2nd ed., Springer, 2004.

Strauss, Walter A. *Partial Differential Equations: An Introduction*. 2nd ed., Wiley, 2008.

Zienkiewicz, O. C., and R. L. Taylor. *The Finite Element Method: Volume 1, The Basis*. 5th ed., Butterworth-Heinemann, 2000.

Academic Papers / Journals

Boyd, John P. “Chebyshev and Fourier Spectral Methods.” Dover Publications, 2001.

Gockenbach, Mark S. “Understanding the Finite Element Method.” *SIAM Review*, vol. 44, no. 4, 2002, pp. 701–705.

Widder, D. V. “The Inversion of the Laplace Integral and the Related Moment Problem.” *American Journal of Mathematics*, vol. 58, no. 1, 1936, pp. 141–160.

Websites and Online Resources

“Fourier Series – An Intuitive Introduction.” BetterExplained, <https://betterexplained.com/articles/an-intuitive-guide-to-fourier-analysis/>. Accessed 9 Oct. 2024.

“Heat Equation.” Wolfram MathWorld, <https://mathworld.wolfram.com/HeatEquation.html>. Accessed 2 Nov. 2024.

“Introduction to the Variational Method.” MIT OpenCourseWare, <https://ocw.mit.edu/courses/mathematics/18-303-linear-partial-differential-equations-fall-2006/lecture-notes/>. Accessed 15 Jan. 2025.

“Laplace Transform.” Paul’s Online Math Notes, <https://tutorial.math.lamar.edu/Classes/DE/LaplaceTransforms.aspx>. Accessed 3 Dec. 2024.

“Partial Differential Equations.” Khan Academy, <https://www.khanacademy.org/math/differential-equations/pde-topic>. Accessed 19 Nov. 2024.

“Talbot’s Method for Numerical Laplace Inversion.” Numerical Recipes Online, <https://numerical.recipes/inversion-laplace-talbot/>. Accessed 4 Jan. 2025.

Zhou, Ting. “Green’s Functions and Their Applications in Physics.” MIT Department of Physics, <https://web.mit.edu/8.13/www/JLExperiments/JLExp10.pdf>. Accessed 29 Oct. 2024.

APPENDIX

Python code written by the author for Graph 6.0.

Python

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
S0 = 1
alpha = 0.1
L = 1
N = 10

# Function definition
def u(x, t, N):
    term1 = (S0 / alpha) * ((L / np.pi) ** 2) * np.sin(np.pi * x / L)
    term2 = sum(
        (np.exp(-alpha * (n * np.pi / L) ** 2 * t) * np.sin(n * np.pi * x / L))
        for n in range(1, N + 1)
    )
    return term1 + term2

# Grid for x and t
x_vals = np.linspace(0, L, 200)
t_vals = np.linspace(0.01, 0.7, 200)
X, T = np.meshgrid(x_vals, t_vals)

# Evaluate u(x, t)
U = np.array([[u(x, t, N) for x in x_vals] for t in t_vals])

# Plot heatmap with contour lines
plt.figure(figsize=(12, 6))
cp = plt.contourf(X, T, U, levels=50, cmap='viridis')
contours = plt.contour(X, T, U, levels=15, colors='black', linewidths=0.5)
plt.clabel(contours, inline=True, fontsize=8)

# Labels and title
plt.colorbar(cp, label='u(x,t)')
plt.xlabel('x')
plt.ylabel('t')
plt.title('Heatmap and Contour Plot of u(x,t)')
plt.grid(False)
plt.tight_layout()
plt.show()
```


Python code written by the author for Graph 7.0.

Python

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Parameters
S0 = 1
alpha = 0.1
L = 1
N = 10

# Define the function
def u(x, t, N):
    term1 = (S0 / alpha) * ((L / np.pi) ** 2) * np.sin(np.pi * x / L)
    term2 = sum(
        (np.exp(-alpha * (n * np.pi / L) ** 2 * t) * np.sin(n * np.pi * x / L))
        for n in range(1, N+1)
    )
    return term1 + term2

# Mesh grid for x and t
x_vals = np.linspace(0, L, 100)
t_vals = np.linspace(0.01, 0.7, 100)
X, T = np.meshgrid(x_vals, t_vals)

# Compute U values
U = np.array([[u(x, t, N) for x in x_vals] for t in t_vals])

# Plot 3D surface
fig = plt.figure(figsize=(12, 8))
ax = fig.add_subplot(111, projection='3d')
surf = ax.plot_surface(X, T, U, cmap='viridis')

# Labels and title
ax.set_xlabel('x')
ax.set_ylabel('t')
ax.set_zlabel('u(x,t)')
ax.set_title('3D Surface Plot of u(x,t)')
fig.colorbar(surf, ax=ax, shrink=0.5, aspect=10)

plt.show()
```